LECTURE

Introduction to IT Ethics and Law

Ethics, morality, law and professional conduct in the information age

This Lecture Slide is prepared with AI assistance (Claude) and reviewed by EKD

LEARNING OUTCOMES

By the end of this lecture you can

1. Define ethics, morality and law, and explain how they differ
2. Explain why professional ethics matters in IT work
3. Analyse an ethical dilemma with a structured decision model
4. Describe how technology, society and law shape each other
5. Apply the IEEE codes of ethics and conduct to a case

Reading: the Chapter 1 and 2 on ethics and IT laws, and the IEEE and IEEE-CS codes (links on the last slide).

FOUNDATIONS

Why start with information systems?

ICT

Computing, multimedia, internet and telecommunication techniques that let people store, process, produce and share information.

Information system

Software, hardware and networks working together to manage data, support decisions and run daily operations.

Systems that hold data and shape decisions carry ethical and legal responsibility.

FOUNDATIONS

Ethics and morality

Shared · reasoned · professional

Ethics

A systematic set of principles a profession or community uses to judge right and wrong.

Example: a code of ethics for engineers

Personal · inherited · cultural

Morality

An individual's or culture's beliefs about right and wrong, shaped by upbringing and experience.

Example: a personal conviction about honesty

Morality is what I believe. Ethics is what we agree to uphold in this role.

FOUNDATIONS

Morality, ethics, and law compared

Aspect	Morality	Ethics	Law
Source	Personal beliefs, culture	Profession or community principles	State authorities
Enforced by	Conscience, social pressure	Peers and professional bodies	Courts and legal sanctions
Change	Slow, personal	Reviewed by the profession	Formal amendment or repeal
Scope	One person	Members of a field	Everyone in the jurisdiction

FOUNDATIONS

Four features of a legal rule

General

Applies to everyone, which guards against arbitrary treatment. It can still give extra protection to some groups.

Mandatory

Breaking it leads to sanctions. Without that, it would be a recommendation, not a rule.

Permanent

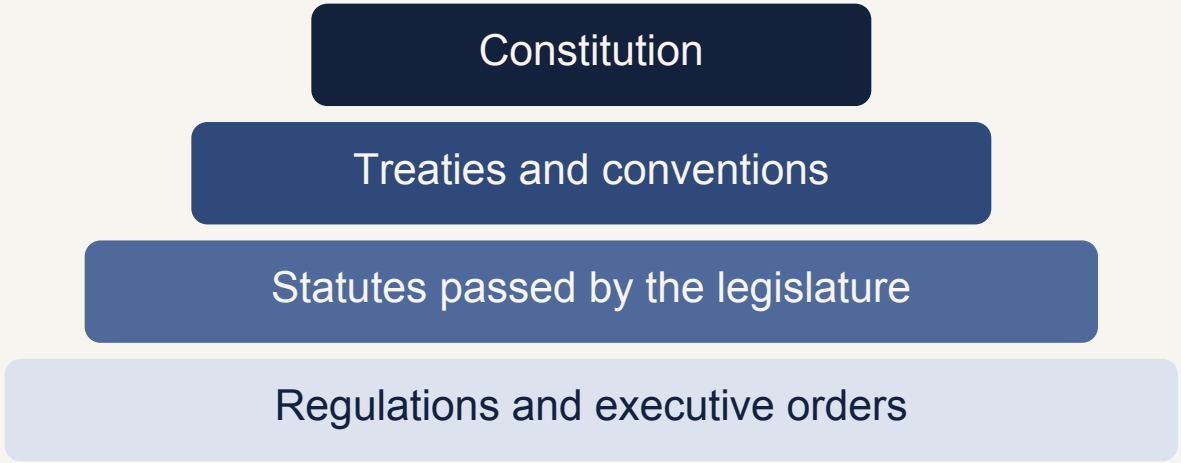
Applies consistently to every situation it covers until the competent authority repeals it. Lasting, not eternal.

Of state origin

Comes from state bodies, and legal rules are ranked in a hierarchy of norms.

FOUNDATIONS

Hierarchy of norms and the law of ICT



Higher norms override lower ones. The exact ranking depends on the country.

Law of ICT

All legislative and case-law provisions that relate to information and communication technologies.

Typical areas: privacy, cybercrime, intellectual property, e-commerce.

FOUNDATIONS

Legal is not always ethical

LEGAL AND ETHICAL

Collecting data with clear, informed consent and a readable privacy notice.

LEGAL BUT UNETHICAL

Using a loophole to track users without meaningful consent.

ILLEGAL BUT ARGUABLY ETHICAL

Breaking a rule to prevent serious harm. Rare, and it needs careful justification.

ILLEGAL AND UNETHICAL

Selling stolen personal data or spreading malware.

ACTIVITY AND RECAP

Think, pair, share

Discuss

- Name one IT action that is legal but unethical.
- Who decides what is ethical in a profession?
- Why can law alone not protect users?

Recap

- Ethics is shared and reasoned; morality is personal.
- Law is general, mandatory, permanent and state-made.
- The three overlap but never fully coincide.

Professional conduct and why ethics matters in IT

Professions, risks, ethical issues and the role of codes

What makes an occupation a profession?

PROFESSIONAL CONDUCT

A profession earns public trust by promising competence, integrity and care for those its work affects.

- Specialised knowledge that others lack
- A duty to the public, not only to the client
- A shared code of ethics and conduct
- Accountability when things go wrong

PROFESSIONAL CONDUCT

Why ethics matters in IT

Scale

One design decision can reach millions of users at once.

Invisibility

Users cannot inspect code, data flows or algorithms, so they must trust the builders.

Knowledge gap

Developers know far more than the people their systems affect.

Permanence

Data can be copied and kept long after it was collected.

PROFESSIONAL CONDUCT

When IT ethics fails: four well-known cases

Bangladesh Government Website — Exposure of Citizens' Personal Data (2023)

City Bank — Customer Financial Data Breach (2024–2025)

Bangladesh Bank Cyber Heist — Theft of $81 Million (2016)

Bangladesh National ID (NID) Database — Alleged Commercial Sale of Citizens' Personal Information (2024)

PROFESSIONAL CONDUCT

Core ethical issues in IT

Privacy
Who may collect, keep and use personal data?

Security
How much protection do users deserve, and who pays?

Intellectual property
Credit, licences, piracy and the open-source commons.

Accuracy
Errors, bias and false claims in data and software.

Access and fairness
Who gets the benefits of technology, and who is left out?

Accountability
Who answers when automated systems cause harm?

PROFESSIONAL CONDUCT

Why professions write codes/rules

Guide

Gives teams an ethical stance to appeal to in hard cases.

Protect

Defines what is improper to ask of an engineer or a team.

Educate

Teaches the public and newcomers what the profession expects.

Codes of ethics guide professionals in making decisions, but professionals must still use their own judgment to decide what is right .

ACTIVITY AND RECAP

Case clinic

Pick one case

- Who was affected, and who had the least power?
- Which ethical issue from the map was at stake?
- What should the engineers have done?

Recap

- A profession earns trust through competence and integrity.
- IT risks grow with scale, opacity and permanence.
- Codes guide judgment but never replace it.

Ethical dilemmas: frameworks and cases

Theories, a decision model and three IT scenarios

ETHICAL DILEMMAS

What is an ethical dilemma?

DEFINITION

A situation where every option has a moral cost, because duties, interests or values conflict.

Typical IT conflicts

- Employer’s wishes versus public safety
- Client confidentiality versus disclosure
- Deadline versus quality
- Profit versus privacy

ETHICAL DILEMMAS

Four lenses for ethical reasoning

Lens	Core question	IT example
Utilitarian	Which choice does the most good for the most people?	Delay a launch to fix a flaw that could harm many users
Deontology	Which duty or rule must I honour whatever the outcome?	Never deceive users about how their data is used
Virtue ethics	What would a person of integrity do here?	Report your own bug promptly instead of hiding it
Rights-based	Whose rights are affected, and are they respected?	Protect privacy even when the data would be useful

ETHICAL DILEMMAS

A six-step decision model

STEP 1

Identify facts

What happened, and what is really being asked?

STEP 2

Map stakeholders

Who is affected, and who has the least power?

STEP 3

Generate options

List at least three realistic courses of action.

STEP 4

Test options

Check each against the law, the code and a lens.

STEP 5

Decide

Choose, and be ready to explain why in public.

STEP 6

Review

Check the outcome and what to do differently.

CASE 1

Ship on schedule, or raise the alarm?

SCENARIO

Your team lead wants to ship a navigation app update on schedule. Testing shows a rare fault that could send users onto unsafe routes. Fixing it would delay the release by a month.

Discuss

- Who could be harmed?
- What do the law and the codes expect?
- Whom do you tell, and in what order?

CASE 2

Data collected for one purpose

SCENARIO

A fitness app’s data reveals when users are away from home. A marketing manager proposes selling this insight to advertisers. The privacy policy is vague on the point.

Discuss

- Is it legal? Is it ethical? Are those the same question?
- Whose consent matters here?
- What would you propose instead?

CASE 3

A gift and a flattering estimate

SCENARIO

You evaluate vendor software for your employer, and the vendor offers you an expensive gift. Your manager also wants an optimistic delivery estimate to win a client.

Discuss

- Which conflicts of interest exist?
- What does an honest estimate require?
- When does a gift become a bribe?

ACTIVITY AND RECAP

Group decision exercise

In groups of four

- Choose one of the three cases.
- Run the six-step model.
- Present a two-minute recommendation.
- Name the code clause that settles it.

Recap

- A dilemma pits duties or values against each other.
- Four lenses give different views of one case.
- A model keeps the reasoning honest and visible.

Technology, society and law

And the professional codes that connect them

TECHNOLOGY, SOCIETY, LAW

A loop, not a line

Technology

Creates new capabilities and new risks.

Example: cloud, AI, drones

Society

Adopts it, benefits from it, and reacts to harm.

Example: privacy concerns, digital divide

Law

Responds with rules that shape design and use.

Example: data protection, cybercrime acts

The loop repeats: new rules change what gets built, and new products raise new questions.

TECHNOLOGY, SOCIETY, LAW

Law moves slower than technology

THE GAP

Products change in months, while legislation takes years. In between, ethics has to do the work.

- Professional codes set shared expectations
- Organisational policies turn them into routine
- Individual judgment handles what rules miss

TECHNOLOGY, SOCIETY, LAW

Main areas of IT law

Data protection

Rules for collecting and sharing personal data.

Cybercrime

Offences such as hacking, fraud and malware.

Intellectual property

Copyright, patents, trademarks and software licences.

Electronic commerce

Online contracts, signatures and payments.

Professional liability

Responsibility for defective or harmful software.

Online content

Speech, defamation, harassment and platform duties.

IEEE CODE OF ETHICS

The IEEE Code of Ethics in three parts

PART 1

Uphold integrity

- Public safety, health and welfare come first
- Protect privacy; practise ethical, sustainable design
- Disclose conflicts of interest; reject bribery
- Be honest and realistic in claims

PART 2

Treat people fairly

- No harassment, including bullying
- No discrimination
- Avoid harming others by false or malicious acts

PART 3

Uphold it together

- Support colleagues in following the code
- Never retaliate against those who report

Summarised from the IEEE Code of Ethics (IEEE Policy 7.8) as published on IEEE society pages; check ieee.org for the current text.

IEEE CODE OF ETHICS

Reading the code in practice

Commitment	What it looks like in IT work
Public welfare and privacy	Run a privacy review before launch
Conflicts of interest	Declare vendor ties before evaluating
Honest claims	Give estimates with their uncertainty
Honest criticism	Accept code review; credit others
Fair treatment	Challenge biased data and design
No retaliation	Protect colleagues who report concerns

IEEE CODE OF CONDUCT

The IEEE Code of Conduct

What participants commit to

- No harassment of any kind, in person or online
- No bullying and no discrimination
- No retaliation against people who report misconduct
- Compliance with applicable laws and IEEE procedures

If something goes wrong

- Report to event staff, venue security or IEEE’s event-conduct contact
- Violations can be investigated and sanctioned, including removal from an event

Summarised from IEEE policy text as quoted on IEEE conference pages; check ieee.org for the current wording.

ACM/IEEE-CS SE CODE

Eight principles for software engineers

Principle	Commitment
1. Public	Act consistently with the public interest
2. Client and employer	Serve their interests, consistent with the public interest
3. Product	Meet the highest professional standards possible
4. Judgment	Keep integrity and independence in professional judgment
5. Management	Managers promote an ethical approach to development
6. Profession	Advance the integrity and reputation of the profession
7. Colleagues	Be fair to and supportive of colleagues
8. Self	Learn throughout life and promote ethical practice

Source: Software Engineering Code of Ethics and Professional Practice, IEEE Computer Society and ACM (computer.org)

ACM/IEEE-CS SE CODE

Using the code well

CENTRAL IDEA

The public interest comes first.

The code is not an algorithm.

Questions to ask

1. Who is affected by this work?
2. Are people treated with due respect?
3. How would a well-informed public judge it?
4. How are the least empowered affected?
5. Would an ideal professional call it worthy?

Source: Software Engineering Code of Ethics and Professional Practice, preamble (computer.org)

PUTTING IT TOGETHER

Codes and law compared

Document	Who it covers	Focus
IEEE Code of Ethics	All IEEE members	Integrity, fairness, upholding the code
IEEE Code of Conduct	IEEE staff, members, event participants	Respectful behaviour: no harassment or retaliation
ACM/IEEE-CS SE Code	Engineers, educators, managers, students	Eight principles of professional practice
Law	Everyone in the jurisdiction	Mandatory rules backed by state sanctions

WRAP-UP

Key takeaways

- Ethics is shared, morality personal, law mandatory.
- Professional ethics protects the public where law is silent or slow.
- Dilemmas need a method: facts, stakeholders, options, tests, decision, review.
- Codes guide judgment. They do not replace it.
- The IEEE and IEEE-CS codes set the baseline for IT professionals.

FOLLOW-UP

Follow-up work and readings

Readings

- Chapter 1 and 2
- IEEE Code of Ethics and Code of Conduct
- ACM/IEEE-CS Software Engineering Code
- IEEE Code of Conduct (ieee.org)